Multiobjective A* Search with Consistent Heuristics

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Abstract. The article describes and analyzes NAMOA*, an algorithm for multiobjective heuristic graph search problems. The algorithm is presented as an extension of A*, an admissible scalar shortest path algorithm. Under consistent heuristics A* is known to improve its efficiency with more informed heuristics, and to be optimal over the class of admissible algorithms in terms of the set of expanded nodes and the number of node expansions. Equivalent beneficial properties are shown to prevail in the new algorithm.

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1. *Introduction*

The multiobjective search problem is an extension of the shortest path problem where arc costs become vectors. This is a genuinely different problem since vector costs induce only a partial order relation.

Hansen [1979] studied the extension of Dijkstra's algorithm [Dijkstra 1959] to the multiobjective case. Loui [1983] further analyzed the problem and showed that some stochastic search problems can be reduced to multiobjective ones. Stewart and White [1991] outlined MOA*, a multiobjective extension of A* [Hart et al. 1968], and presented proofs on admissibility, node expansion, and comparison of heuristics. Dasgupta et al. [1999] extended this work in several ways, including versions for non-consistent heuristics (MOA**) and limited memory

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(MOMA*), with applications in the area of VLSI design and the log cutting problem. Perny and Spanjaard [2002, 2005] presented a generalization of MOA* with application to a Web access problem. Mandow and Pérez de la Cruz [2003] presented a systematic extension of the heuristic search paradigm to the multicriteria case, and analyzed admissibility conditions for algorithms with different multicriteria preference relations (multiobjective, multiattribute, lexicographic, and goal-based). Refanidis and Vlahavas [2003] described an application of heuristic search with a multiattribute preference to domain-independent planning. Other interesting applications of multiobjective search include route planning [Alechina and Logan 2001] and satellite scheduling [Gabrel and Vanderpooten 2002].

Mandow and Pérez de la Cruz [2005] reconsidered the extension of A* to the multiobjective case, outlined a new algorithm (NAMOA* - New Approach to Multi-Objective A*), and briefly discussed its admissibility. The basic operations in A* are selection and expansion of open nodes at each iteration. In A*, each open node stands for a single partial solution path that can be further expanded. However, as explained in the paper, this is no longer the case in multiobjective problems, where many interesting paths may reach the same node simultaneously. The MOA* algorithm [Stewart and White 1991], and subsequent extensions, were devised preserving *node* selection and expansion as the algorithm's basic operations. On the other hand, NAMOA* preserves *path* selection and expansion as the basic operations. Experimental tests showed a reduction in the space requirements of NAMOA* over those of MOA*. Further analysis [Mandow and Pérez de la Cruz 2006] revealed that the new algorithm also shows important properties, analogous to those of A*, that are not shared by MOA*. In particular, better informed consistent heuristics always ensure equally or more efficient search.

This article presents NAMOA* in full detail, introduces a complete and detailed proof on its admissibility, and extends the analysis on the efficiency of heuristics to show that, under consistent heuristics, the algorithm can also be proven optimal over the class of admissible multiobjective search algorithms. These results are compared to those previously obtained for MOA* [Stewart and White 1991].

The article is organized as follows: Section 2 reviews previous relevant work in scalar search and points out analogies and differences with the multiobjective search problem. Section 3 presents the new algorithm and illustrates its behaviour with a simple example. Relevant differences with MOA* are identified. Section 4 proves the admissibility of the new algorithm. Section 5 considers the effect of heuristic information on the efficiency of the new algorithm. Optimality under consistent heuristics is established in Section 6. Finally, conclusions and future research are briefly summarized.

2. *Scalar and Multiobjective Search*

2.1. PREVIOUS RESULTS IN SCALAR GRAPH SEARCH. The shortest path problem can be stated as follows: Let G be a locally finite labeled directed graph $G = (N, A, c)$, of $|N|$ nodes, and $|A|$ arcs (n, n') labeled with positive costs $c(n, n') \in \mathbb{R}$. Let a path in G be any sequence of nodes $P = (n_1, n_2, \ldots n_k)$ such that for all $i < k$, $(n_i, n_{i+1}) \in A$, and the cost of each path $c(P)$ be the sum of the costs of its component arcs. Given a start node $s \in N$, and a set of goal nodes $\Gamma \subseteq N$, find the minimum cost path in G from s to a node in Γ.

The A* algorithm [Hart et al. 1968] is an efficient solution to this problem. It is a best-first search algorithm that uses a search tree to store the set of known interesting partial solution paths, a list of open nodes that can be further expanded, and a characteristic evaluation function $f(n) = g(n) + h(n)$ to select the next open node to be expanded. While function $g(n)$ denotes the cost of the path stored in the tree from s to n, the heuristic function $h(n)$ estimates the cost of a solution from node n to a goal node. Heuristic functions can be devised for particular problems depending on their structure.

Several important results have been presented to analyse the relationship between the properties of $h(n)$ and those of A*,

Admissibility. Let $h^*(n)$ be the real optimal cost of a path from n to a goal node. When the heuristic is an optimistic estimate ($h(n) \leq h^*(n)\ \forall n$), the search is admissible, that is, if a solution exists it is guaranteed to find an optimal one. Under additional assumptions ($h(n) \geq 0\ \forall n \in N$, and $c(n, n') \geq \epsilon > 0\ \forall (n, n') \in A$) A* is admissible even on infinite graphs.

Efficiency. The efficiency of A* has been conveniently related to the accuracy of $h(n)$. In particular, important results can be proven when $h(n)$ is *consistent* or satisfies the equivalent *monotone* property. Let $k(n, n')$ denote the cost of an optimal path in G from n to n'. A heuristic function $h(n)$ is consistent when,

$$h(n) + k(n, n') \leq h(n') \quad \forall n, n' \in N \tag{1}$$

Alternatively, $h(n)$ is said to be monotone when,

$$h(n) + c(n, n') \leq h(n') \quad \forall (n, n') \in A \tag{2}$$

When $h(n) = 0$ for all nodes, A* behaves like Dijkstra's algorithm. Under consistent $h(n)$ A* solves the problem in $O(|N|)$ iterations, storing $O(|N|)$ nodes in memory, in the worst case. In particular, if c^* denotes the optimal solution cost, A* will always expand all nodes with $f(n) < c^*$, but at most those with $f(n) \leq c^*$. Therefore, given a consistent heuristic estimate, greater values of $h(n)$ can push the evaluation of more and more nodes beyond the $f(n) = c^*$ frontier, reducing search effort.

Optimality. When the heuristic is consistent, A* can also be proven optimal among the class of admissible algorithms both in terms of the set of nodes expanded and the number of node expansions (iterations) [Dechter and Pearl 1985]. This means that no algorithm in this class can skip a node expanded by A* without compromising its admissibility.

2.2. Multiobjective Graph Search

2.2.1. *Problem Definition.* The multiobjective search problem can be stated as follows: Let G be a locally finite labeled directed graph $G = (N, A, \vec{c})$, of $|N|$ nodes, and $|A|$ arcs (n, n') labeled with positive vectors $\vec{c}(n, n') \in \mathbb{R}^q$. Let a path in G be any sequence of nodes $P = (n_1, n_2, \ldots . n_k)$ such that for all $i < k$, $(n_i, n_{i+1}) \in A$, and the cost vector of each path $\vec{c}(P)$ be the sum of the costs vectors of its component arcs. Given a start node $s \in N$, and a set of goal nodes $\Gamma \subseteq N$, find the set of *non-dominated* cost paths in G from s to nodes in Γ.

2.2.2. *Dominance.* In multiobjective problems cost vectors $\vec{c}(n, n')$ induce only a partial order preference relation $\prec$ called *dominance*,

$$\forall \vec{f}, \vec{f'} \in \mathbb{R}^q \quad \vec{f} \prec \vec{f'} \quad \Leftrightarrow \quad \forall i \quad f_i \leq f'_i \ \wedge \ \vec{f} \neq \vec{f'} \tag{3}$$

where f_i denotes the i-th element of vector $\vec{f}$.

Therefore, given two q-dimensional vectors $\vec{f}$ and $\vec{f}'$, it is not always possible to rank one as better than the other. For example, in a two dimensional cost space, vector (2, 3) dominates (2, 4), but no dominance relation exists between vectors (2, 3) and (3, 2).

Given a set of vectors X, we shall define $nondom(X)$ the set of nondominated vectors in set X in the following way,

$$nondom(X) = \{\vec{x} \in X \mid \nexists \vec{y} \in X \quad \vec{y} \prec \vec{x}\} \tag{4}$$

We shall find it useful later on to denote by $\preceq$ the relation "dominates or equals". The relationship "is indifferent to" will be denoted by $\sim$, and defined as $\vec{x} \sim \vec{y} \iff \vec{x} \not\prec \vec{y} \wedge \vec{y} \not\prec \vec{x}$. The relationship "dominates or is indifferent to", will be denoted by $\precsim$, and defined as $\vec{x} \precsim \vec{y} \iff \vec{x} \prec \vec{y} \vee \vec{x} \sim \vec{y}$.

2.2.3. *Extensions of A* to the Multiobjective Case.* The essence of the multiobjective search problem is to find the set of *all* nondominated solution paths. Therefore, an analogy could be traced to a version of A* that aims at finding *all* optimal paths. However, some important differences with the scalar search problem can be pointed out.

First of all, given the dominance preference relation, two (or more) incomparable (non-dominated) paths may reach any given node from the same or different parents. Several important consequences can be identified:

(1) The search tree used by A* to record the best known path to generated nodes is no longer sufficient. A directed acyclic graph can be used instead, to record the set of nondominated known paths to generated nodes. We shall call this data structure a *search graph*.
(2) The number of generated nodes may no longer be a realistic estimate of the memory needed by the algorithm. A variable number of arcs and non-dominated cost vectors reaching each node need to be recorded as well.
(3) Finally, each time a new path is generated to a known node, its cost may need to be tested for dominance with the set of all recorded costs reaching the node. The computational cost of this operation grows with the number of cost vectors stored at each node.

We shall denote by $\vec{g}(P)$ the cost vector of each path stored in the search graph. In general, heuristic estimates will involve a set of vectors $H(n)$ for each node n, estimating cost vectors of paths from n to each goal node. Therefore, for each path P_{sn} from s to n with cost $\vec{g}(P) = \vec{g}_P$, there will be a set of heuristic evaluation vectors, $F(P_{sn})$. This function is the multiobjective analogue to $f(n)$ in A*,

$$F(P_{sn}) = F(n, \vec{g}_P) = nondom\{\vec{f} \mid \vec{f} = \vec{g}_P + \vec{h} \ \wedge \vec{h} \in H(n)\}$$

Finally, note that at each iteration, A* selects and expands an open node, i.e. considers all the possible extensions of the path stored in the search tree to that node. In A*, each open node stands for a single partial solution path that can be

TABLE I. ALGORITHM NAMOA*

(1) CREATE:
—An acyclic search graph *SG* rooted in s.
—Two sets $G_{op}(s) = \{\vec{0}\}$, $G_{cl}(s) = \emptyset$.
—A list of alternatives, $OPEN = \{(s, \vec{g}_s, F(s, \vec{g}_s))\}$.
—Two empty sets, *GOALN*, *COSTS*.

(2) CHECK TERMINATION. If *OPEN* is empty, then search backwards in *SG* from the nodes in *GOALN* taking into account the costs in *COSTS* and return the completelly labelled solution subgraph.

(3) PATH SELECTION. Select an alternative $(n, \vec{g}_n, F)$ from OPEN with $\vec{f} \in F$ non-dominated in *OPEN*, i.e.

$$\forall (n', \vec{g}_{n'}, F') \in OPEN \quad \nexists \vec{f}' \in F' \quad | \quad \vec{f}' \prec \vec{f} \tag{5}$$

Delete $(n, \vec{g}_n, F)$ from *OPEN*, and move $\vec{g}_n$ from $G_{op}(n)$ to $G_{cl}(n)$.

(4) SOLUTION RECORDING. If $n \in \Gamma$, then
—Include n in *GOALN* and $\vec{g}_n$ in *COSTS*.
—For all alternatives $(x, \vec{g}_x, F_x)$ in OPEN, eliminate from F_x all vectors dominated by $\vec{g}_n$.
—Eliminate from OPEN all alternatives $(x, \vec{g}_x, F_x)$ such that F_x is empty.
—Go back to step 2

(5) Path EXPANSION: If $n \notin \Gamma$, then,
For all successor nodes m of n that do not produce cycles in *SG* do:
(a) Calculate the cost of the new path found to m: $\vec{g}_m = \vec{g}_n + \vec{c}(n, m)$.
(b) If m is a new node
  i. Calculate $F_m = F(m, \vec{g}_m)$ filtering estimates dominated by *COSTS*.
  ii. If F_m is not empty, put $(m, \vec{g}_m, F_m)$ in *OPEN*, set $G_{op}(m) = \{\vec{g}_m\}$, and label with $\vec{g}_m$ a pointer to n.
  else if $\vec{g}_m$ equals some cost vector in $G_{op}(m) \cup G_{cl}(m)$, then,
  i. label with $\vec{g}_m$ a pointer to n,
  else if $\vec{g}_m$ is non-dominated by any cost vectors in $G_{op}(m) \cup G_{cl}(m)$ (a path to m with new interesting cost has been found), then,
  i. Eliminate from $G_{op}(m)$ and $G_{cl}(m)$ vectors dominated by $\vec{g}_m$ (pruning). For each vector $\vec{g}'$ eliminated from $G_{op}(m)$, eliminate the corresponding tuple $(m, \vec{g}', F)$ from *OPEN*.
  ii. Calculate $F_m = F(m, \vec{g}_m)$ filtering estimates dominated by *COSTS*.
  iii. If F_m is not empty, put $(m, \vec{g}_m, F_m)$ in *OPEN*, and put $\vec{g}_m$ in $G_{op}(m)$ labelling a pointer to n.
endFor;
Go back to step 2.

further expanded. However, in the multiobjective case, if an acyclic graph is used instead of a tree to record partial solutions, this will no longer be the case. The MOA* algorithm [Stewart and White 1991] was devised preserving *node* selection and expansion as the algorithm's basic operations. The next section describes an extension of A* to the multiobjective case that preserves *path* selection and expansion as the basic operations.

3. *Algorithm NAMOA**

3.1. BRIEF DESCRIPTION. The pseudocode of NAMOA*, an algorithm for multiobjective A* search, is shown in Table I. This is a more detailed and complete version of the algorithm originally presented in Mandow and Pérez de la Cruz [2005].

In accordance with the considerations in Section 2.2, this extension of A* to the multiobjective case is devised in the following way.

—The algorithm uses an acyclic search graph *SG* to record interesting partial solution paths. For each node n in *SG* two sets, $G_{cl}(n)$ and $G_{op}(n)$, denote the sets of nondominated cost vectors of paths reaching n whose extensions have or have not been explored yet respectively (i.e., closed or open). Each cost vector in these sets labels one or more pointers emanating from n to its parents with that cost. Initially, s is the only node in *SG*.

—The algorithm keeps a list *OPEN*, of partial solution paths that can be further expanded. For each node n in *SG* and each nondominated cost vector $\vec{g} \in G_{op}(n)$, there should be a corresponding pair $(n, \vec{g})$ in *OPEN*. However, for the sake of clarity, the set of interesting heuristic evaluation vectors associated to every such pair will be included to form a corresponding triple $(n, \vec{g}, F)$. Initially $F = F(n, \vec{g})$. Therefore, the algorithm will store such triples in *OPEN*. Initially, $(s, \vec{g}_s, F(s, \vec{g}_s))$ is the only triple in *OPEN*.[1]

—At each iteration, the algorithm will consider the extension of an open triple $(n, \vec{g}, F)$ that stands for a partial solution path from s to n with cost $\vec{g}$. This operation is called *path expansion*.

—According to the principle of optimality, any path P stored in *SG* from s to n will be discarded, or *pruned*, whenever there is another path P' reaching n such that $\vec{g}(P') \prec \vec{g}(P)$.

—Two sets, *GOALN* and *COSTS*, record all goal nodes reached and the set of all non-dominated cost vectors found to goal nodes respectively. Once a solution is known its cost vector can be used to discard, or *filter*, heuristic evaluation vectors from $F(P)$ for each open path P. When all heuristic evaluation vectors in $F(P)$ are dominated by cost vectors in *COSTS*, path P is filtered from *OPEN*. From the point when the first solution is found, *COSTS* is also used to filter heuristic evaluation vectors of new paths entering *OPEN*. For any triple $(n, \vec{g}, F)$, the set F is formally defined as the set of cost estimates in $F(n, \vec{g})$ not dominated by vectors in *COSTS*. As new vectors are included in *COSTS*, the sets are updated accordingly.

—The search terminates only when the *OPEN* list is empty. Note that the fate of every path (triple) that enters *OPEN* is to be either selected for expansion, pruned, or filtered.

—Note that the number of distinct optimal solutions paths can grow exponentially in the worst case with the number of nodes in the graph. This is true even for single objective problems. Therefore, it can be infeasible to explicitly enumerate all of them once the *OPEN* list becomes empty.

An alternative is to return the solution set implicitly, that is, the algorithm returns the *solution subgraph*, defined as the subgraph formed only by all arcs and nodes that take part in at least one nondominated solution path. Once the *OPEN* list becomes empty, this solution subgraph can be constructed from the search graph *SG*, searching backwards from the nodes in *GOALN*. All nodes n in the solution subgraph can also be labeled during this backward search with $G^*(n)$ and $H^*(n)$ sets. These sets contain the cost vectors of nondominated paths that go from s to

[1] The Operational Research literature often refers to these pairs (or, in our case, triples) as *labels* or *path tips*. In this article, we refer to them simply as *paths* or *alternatives*.

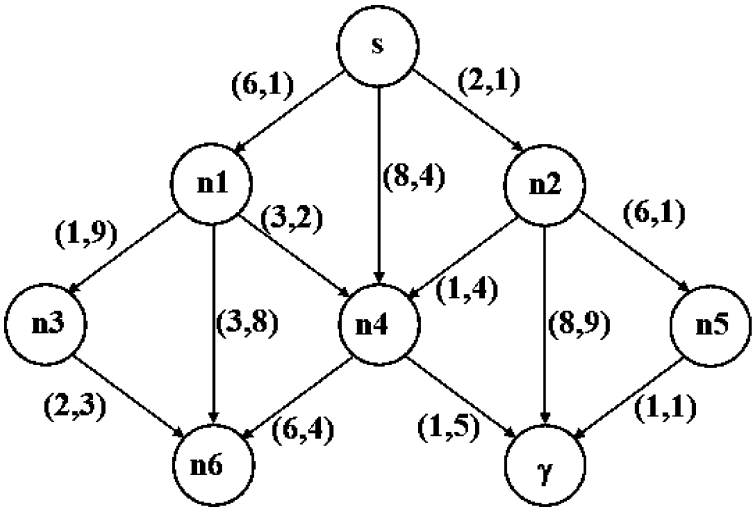


FIG. 1. Sample graph.

TABLE II. HEURISTIC FUNCTION

n	s	n_1	n_2	n_3	n_4	n_5	n_6	γ
H(n)	(3,3)	(2,2)	(2,2)	(1,1)	(1,1)	(1,1)	(0,0)	(0,0)

n, and from n to goal nodes respectively. The solution subgraph can be queried to return information on the set of solutions, for example, a solution with a particular nondominated cost can be obtained without further search, simply traversing the subgraph from the start to the goal (or vice-versa), checking the arc costs and the $G^*(n)$ and $H^*(n)$ sets.

3.2. EXAMPLE. Let us consider the sample graph in Figure 1, where s is the start node, and γ the only goal node. A heuristic evaluation function $H(n)$ is presented in Table II. A single heuristic vector is given for each node for the sake of simplicity. A run of NAMOA* over this problem illustrates pruning and filtering operations, and the updating of the G_{op} and G_{cl} sets. A trace of the search graph is shown in Figures 2 through 7. Values of G_{op} and G_{cl} are shown at each iteration. Values that do not change from the previous iteration are omitted for brevity. Table III shows a trace of *OPEN*.

At iteration 1, *SG* has only node s at its root, $G_{op}(s) = \{(0, 0)\}$, and $G_{cl}(s) = \emptyset$. The only path in *OPEN* is selected, and its three extensions to nodes n_1, n_2, and n_4 are added to *SG* and *OPEN*. At iteration 2, only n_2 has a nondominated estimate in *OPEN*. It is selected, and its extensions to n_4, n_5, and γ are added to *SG* and *OPEN*. Note that two nondominated paths lead now to node n_4 (each cost vector in $G_{op}(n_4)$ would label a different pointer). At iteration 3, two alternatives have nondominated estimates in *OPEN* (n_1 with estimate (8, 3), and n_4 with estimate (4, 6)). Let us assume ties between nondominated estimates are always broken by choosing the smaller first component in estimated cost vectors. The path leading to n_4 would be selected, and its extensions to γ and n_6 generated. The new path reaching γ dominates and prunes the old one, and $G_{op}(\gamma)$ is updated accordingly. At iteration 4, a path leading to the goal node γ has a nondominated estimate in

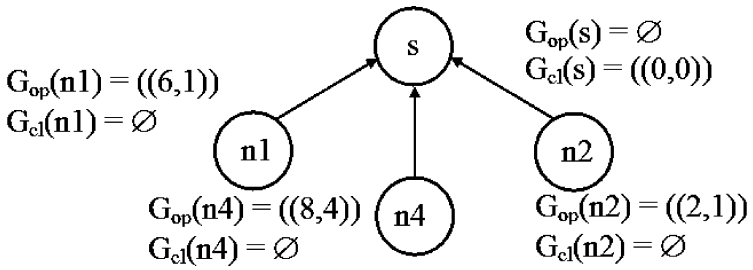


FIG. 2. Search graph (iteration 2).

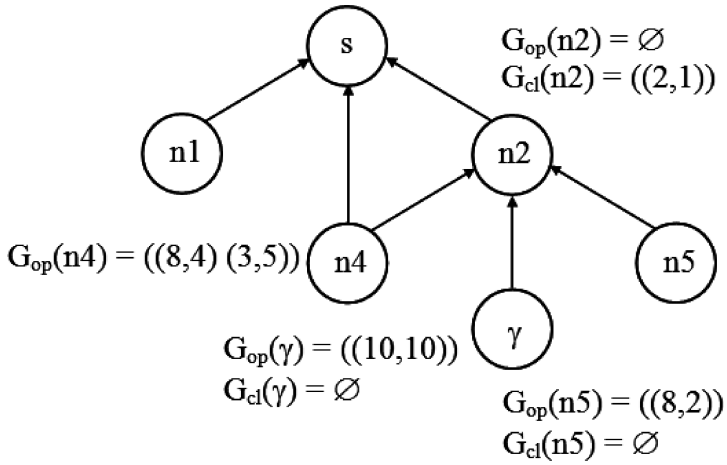


FIG. 3. Search graph (iteration 3).

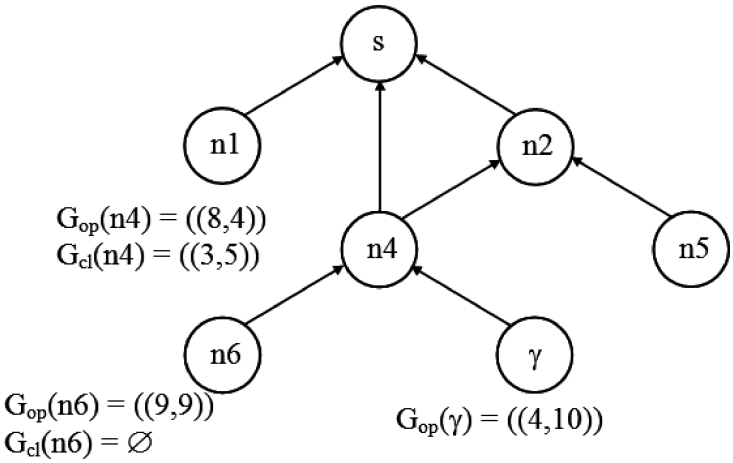


FIG. 4. Search graph (iteration 4).

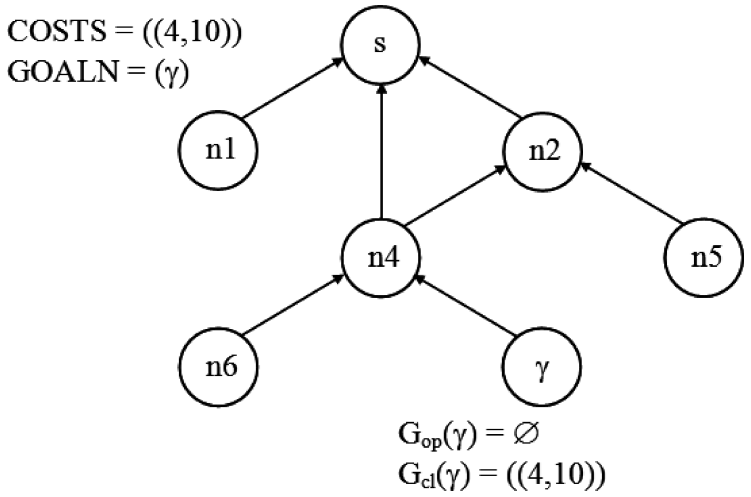


FIG. 5. Search graph (iteration 5).

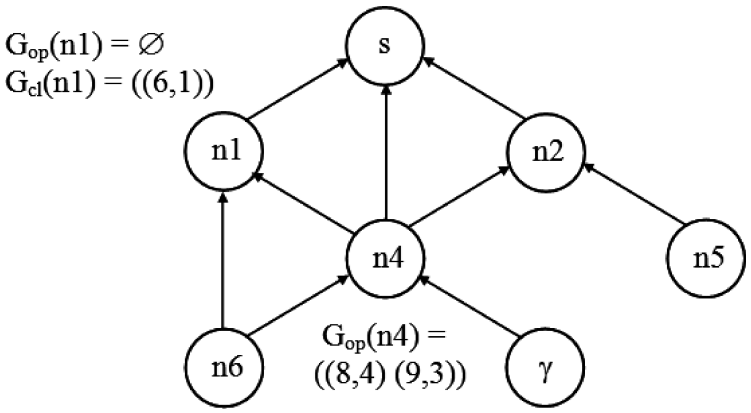


FIG. 6. Search graph (iteration 6).

OPEN. It is selected, added to *GOALN*, and its cost vector included in *COSTS*. At iteration 5, the only nondominated alternative, leading to node n_1, is selected. Three extensions to nodes n_3, n_6, and n_4 are generated with varying fates. The extension to n_3 with estimate (8, 11) is filtered by *COSTS* and never enters the *OPEN* list. The extension to n_6 results in an additional path reaching that node with the same cost of (9, 9) already in $G_{op}(n_6)$. Therefore, $G_{op}(n_6)$ is unchanged, and a new pointer labeled with (9, 9) is set to n_1. Finally, the extension to n_4 generates an additional path to that node with a new nondominated cost of (9, 3). The only nondominated alternatives are selected at iterations 6 and 7. At iteration 7, a new solution path leading to γ is selected. Therefore, the set *COSTS* is updated to {(4, 10)(9, 3)}. At iteration 8, all remaining alternatives are filtered and *OPEN* is empty. The algorithm would search backwards from γ returning the solution subgraph with the two paths found with costs (4, 10) and (9, 3) (Figure 8).

3.3. SELECTION AND EXPANSION: PATHS VERSUS NODES. For more than a decade MOA* has been the reference multiobjective heuristic search algorithm.

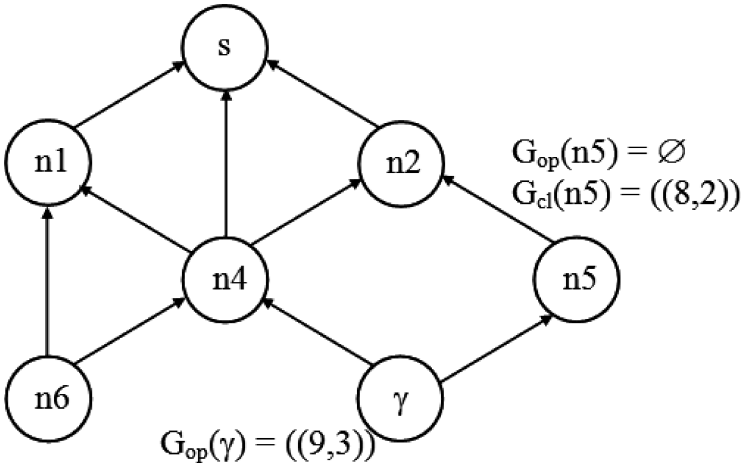


FIG. 7. Search graph (iteration 7).

Unlike the algorithm presented in Section 3.1, MOA* is built around the idea of node selection and expansion. In particular, it makes no distinction in *SG* between the sets $G_{op}(n)$ and $G_{cl}(n)$, and a single set $G(n) = G_{op}(n) \cup G_{cl}(n)$ is kept for each node. A heuristic evaluation function $F(n)$ is defined for each *node*,

$$F(n) = nondom\{\vec{f} \mid \vec{f} = \vec{g} + \vec{h} \ \wedge \vec{g} \in G(n) \wedge \vec{h} \in H(n)\}$$

For the sake of comparison, the *OPEN* list can be considered a list of pairs $(n, F(n))$ where nodes with $\vec{f} \in F(n)$ nondominated in *OPEN* and nondominated by *COSTS* are selected. The distinction between individual paths reaching each node n is lost and, accordingly, all known paths reaching each node n are either simultaneously open or closed. Essentially, MOA* presents two distinguishing features when compared to NAMOA*.

(1) Each time a new nondominated path is found to a closed node, the whole node is put back into *OPEN*.
(2) Each time a node is selected for expansion, all known nondominated paths reaching that node are extended simultaneously.

The first difference may result in unnecessary reexpansion of nodes. The second may result in unnecessary extension of paths in *SG* and storage of uninteresting cost vectors. Additionally, when a goal node γ is selected, all cost vectors in $G(\gamma)$ may enter in *COSTS*. In general, it is not possible to know which of these cost vectors belong to truly non-dominated solutions until all of them have been found.

These differences can be illustrated with the example presented in Section 3.2. Table IV shows a trace of OPEN for MOA*. Provided ties in the selection of OPEN nodes were broken in the same way, a search with MOA* would produce the same results in the first two iterations. In iteration 3, the selection of node n_4 for expansion would produce in MOA* the extension of *both* known non-dominated paths to that node. Accordingly, all cost vectors in $G(n_4) = \{(8, 4)(3, 5)\}$ would be extended resulting in $G(n_6) = \{(14, 8)(9, 9)\}$ and $G(\gamma) = \{(9, 9)(4, 10)\}$, and node n_4 would be marked as *closed*. Note that NAMOA* did not need to store cost vectors (14, 8) in node n_6, nor (9, 9) in γ.

TABLE III. *OPEN* TUPLES AT EACH ITERATION FOR NAMOA* (dom = DOMINATED; ←= SELECTED)

It.	*OPEN*
1	$(s, (0, 0), ((3, 3)))$ ←
2	$(n_1, (6, 1), ((8, 3)))$ dom $(n_2, (2, 1), ((4, 3)))$ ← $(n_4, (8, 4), ((9, 5)))$ dom
3	$(n_1, (6, 1), ((8, 3)))$ $(n_4, (8, 4), ((9, 5)))$ dom $(n_4, (3, 5), ((4, 6)))$ ← $(n_5, (8, 2), ((9, 3)))$ dom $(\gamma, (10, 10), ((10, 10)))$ dom
4	$(n_1, (6, 1), ((8, 3)))$ $(n_4, (8, 4), ((9, 5)))$ dom $(n_5, (8, 2), ((9, 3)))$ dom $(\gamma, (4, 10), ((4, 10)))$ ← $(n_6, (9, 9), ((9, 9)))$ dom
5	$(n_1, (6, 1), ((8, 3)))$ ← $(n_4, (8, 4), ((9, 5)))$ dom $(n_5, (8, 2), ((9, 3)))$ dom $(n_6, (9, 9), ((9, 9)))$ dom
6	$(n_4, (8, 4), ((9, 5)))$ dom $(n_5, (8, 2), ((9, 3)))$ ← $(n_6, (9, 9), ((9, 9)))$ dom $(n_4, (9, 3), ((10, 4)))$ dom
7	$(n_4, (8, 4), ((9, 5)))$ dom $(n_6, (9, 9), ((9, 9)))$ dom $(n_4, (9, 3), ((10, 4)))$ dom $(\gamma, (9, 3), ((9, 3)))$ ←

At iteration 4, node γ would be selected, and two different cost vectors $\{(9, 9)(4, 10)\}$ would be included in *COSTS*. One of them does not belong to a truly nondominated solution, but MOA* will not be able to establish this until termination.

Finally, note that at iteration 5, the expansion of node n_1 results in a new nondominated path to n_4. MOA* would place accordingly n_4 back in *OPEN* with *all* its heuristic evaluation cost vectors. This implies that prior to termination, MOA* will need to reexpand n_4. This was unnecessary in NAMOA* since the newly found path to n_4 can never lead to a nondominated solution.

These considerations suggest that MOA*, a provably admissible algorithm, may perform unnecessary operations. The next section shows that NAMOA* is also admissible and that these extra operations can be safely skipped.

4. *Admissibility of NAMOA**

This section presents a complete proof on the admissibility of the algorithm, and bounds on the number of paths expanded and stored in memory.

A scalar algorithm is said to be *admissible* if it is guaranteed to return an optimal solution whenever a solution exists. We extend the definition as follows: a multiobjective search algorithm is *admissible* if it terminates with the set of *all* nondominated solutions whenever this set is finite and nonempty, or if it does not terminate whenever this set is infinite.

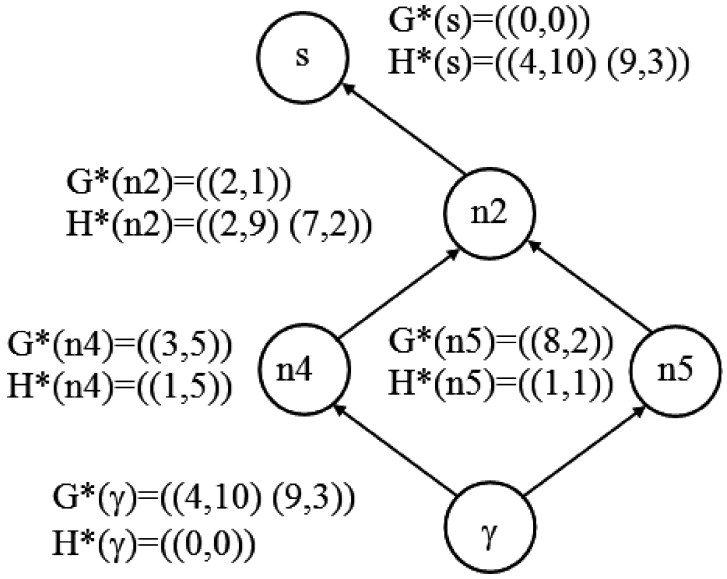


FIG. 8. Solution subgraph.

TABLE IV. VALUES OF $G(n)$ AND $F(n)$ OF *OPEN* NODES AT ITERATIONS 1-6 FOR MOA* (NONDOMINATED VECTORS IN BOLD FACE; ←= SELECTED)

It.	n	$G(n)$	$F(n)$
1	s ←	((0, 0))	((**3,3**)))
2	n_1	((6, 1))	((8, 3)))
	n_2 ←	((2, 1))	((**4,3**)))
	n_4	((8, 4))	((9, 5)))
3	n_1	((6, 1))	((**8,3**)))
	n_4 ←	((8, 4)(3, 5))	((9, 5)(**4,6**)))
	n_5	((8, 2))	((9, 3)))
	γ	((10, 10))	((10, 10)))
4	n_1	((6, 1))	((**8,3**)))
	n_5	((8, 2))	((9, 3)))
	γ ←	((9, 9)(4, 10))	((9, 9)(**4,10**))
	n_6	((14, 8)(9, 9))	((14, 8)(9, 9))
5	n_1 ←	((6, 1))	((**8,3**)))
	n_5	((8, 2))	((9, 3)))
	n_6	((14, 8)(9, 9))	((14, 8)(9, 9))
6	n_5	((8, 2))	((**9,3**)))
	n_6	((14, 8)(9, 9))	((14, 8)(9, 9))
	n_4 ←	((8, 4)(3, 5)(9, 3))	(9, 5)((**4,6**))(10, 4))
...	...		

The proofs presented in this section rely on a set of reasonable assumptions, analogous to those presented in Stewart and White [1991] to prove the admissibility of MOA*:

(1) The graph $G = (N, A)$ to be searched is locally finite.
(2) The heuristic function $H(n)$ is admissible, that is, for all nondominated solution paths $P^* = (s, n_1, \ldots, n_i, n_{i+1} \ldots \gamma_k)$, $\gamma_k \in \Gamma$ and each subpath

$P_i^* = (s, n_1, \ldots, n_i)$ of P^*,

$$\exists \vec{h} \in H(n_i) \mid \vec{g}(P_i^*) + \vec{h} \preceq \vec{g}(P^*)$$

(3) For infinite graphs,
   (a) all components of heuristic vectors are nonnegative, that is, $h_k \geq 0 \quad \forall k \; \forall n \; \forall \vec{h} \in H(n)$
   (b) all components of cost vectors have a positive lower bound, that is, $c_k(n, n') \geq \epsilon > 0 \quad \forall k \; \forall (n, n') \in A$

We shall find it useful to denote by C^* the set of all non-dominated solution costs, and by $\vec{c}^*$ some member of C^*.

LEMMA 4.1. *Let $P^* = (s, n_1, \ldots, n_i, n_{i+1} \ldots \gamma)$ be some nondominated solution path with cost $\vec{g}(P^*) = \vec{c}^*$. Whenever some subpath $P_i^* = (s, n_1, \ldots, n_i)$ of P^* is in OPEN (i.e., $\vec{g}(P_i^*) \in G_{op}(n_i)$), there exists some $\vec{f}$ in $F(P_i^*)$ such that $\vec{f} \preceq \vec{c}^*$.*

PROOF. The lemma follows trivially from the definition of F (see Section 2.2.3), the definition of admissible heuristic (Assumption (2)), and the fact that, by definition, such $\vec{f}$ will never be filtered. □

THEOREM 4.2. *For each nondominated solution path $P^* = (s, n_1, \ldots, n_i, n_{i+1} \ldots \gamma)$ with cost $\vec{g}(P^*) = \vec{c}^*$, there is always before its discovery a subpath $P_i^* = (s, n_1, \ldots, n_i)$ of P^* such that: a) P_i^* is recorded in SG; b) $\vec{g}(P_i^*) \in G_{op}(n_i)$.*

PROOF. The theorem can be proven by induction on the number of iterations. The proposition is true for all solutions at iteration 1, when s is at the root of *SG*, and $(s, \vec{0}, H(s))$ is the only tuple in *OPEN*.

Let P_i^* make the proposition true for some nondominated solution P^* at some iteration k. Recall that the fate of every path that enters *OPEN* is to be either selected, pruned, or filtered. By definition, P_i^* is nondominated to n_i, therefore $g(P_i^*)$ will never be pruned from $G_{op}(n_i)$. From Lemma 4.1 P_i^* will never be filtered either. In consequence, P_i^* will remain in *OPEN* satisfying the theorem for P^* until it is eventually selected for expansion.

Let us assume P_i^* is selected for expansion at iteration $j > k$. If $n_i = \gamma$, then $P_i^* = P^*$ is discovered. Otherwise, $n_i \neq \gamma$, a new nondominated path to n_{i+1}, $P_{i+1}^* = (s, n_1, \ldots, n_i, n_{i+1})$ will be generated, $g(P_{i+1}^*)$ included in $G_{op}(n_{i+1})$, and P_{i+1}^* will satisfy the proposition for P^*. □

THEOREM 4.3. *If there is a solution, the algorithm terminates even on infinite graphs.*

PROOF. All best first search algorithms that prune cycles terminate on finite graphs [Pearl 1984]. For an infinite graph let us assume there is at least a nondominated solution P^* with finite cost $\vec{c}^* = (c_1^*, c_2^*, \ldots, c_q^*)$, and let $c_m = \max_i\{c_i^*\}$. From Theorem 4.2, some subpath P_i^* of P^* is always in *OPEN* before its discovery, and from Lemma 4.1 there exists $\vec{f} \in F(P_i^*)$ such that $\vec{f} \preceq \vec{c}^*$. From Assumption (3)(b), every path P with length longer than c_m/ϵ arcs will trivially attain $\vec{g}(P)$ such that $\vec{c}^* \prec \vec{g}(P)$. From assumption 3a, all heuristic vectors are nonnegative, and therefore for every such path $P, \vec{c}^* \prec \vec{f} \quad \forall \vec{f} \in F(P)$. Since the graph G is locally finite (assumption 1), there can only be a finite number of paths

with length shorter or equal to c_m/ϵ, and hence nondominated by $\vec{c}^*$. Now, in the worst case, all paths with some estimate nondominated by $\vec{c}^*$ (including subpaths of P^*) will be selected in a finite number of steps. Then, P^* will be found and selected, filtering all remaining paths and leaving *OPEN* empty. □

COROLLARY 4.4. *Whenever there is a solution, the set of nondominated solutions must be finite.*

PROOF. Let P be a solution with cost $\vec{c}$. All nondominated solutions must be nondominated by $\vec{c}$. The proof of Theorem 4.3 shows the set of such candidates must be finite. □

LEMMA 4.5. *Each path $P = (s, \ldots, n)$ selected from OPEN for expansion satisfies upon selection that,*

$$\exists \vec{h} \in H(n) \quad | \quad \nexists \vec{c}^* \in C^*, \quad \vec{c}^* \prec \vec{g}(P) + \vec{h}$$

PROOF. Let us assume, for the purpose of contradiction, that a path P is selected and that for all heuristic evaluation vector $\vec{f}' \in F(P)$ exists $\vec{c}^* \in C^* \mid \vec{c}^* \prec \vec{f}'$. There must be at least one such vector $\vec{c}^* \notin COSTS$, as otherwise all the evaluation vectors would have been filtered. Now, from Theorem 4.2, for all $\vec{c}^* \in C^*$ not yet found there is at least one path $P_i = (s, \ldots, n_i)$ with $\vec{f} \in F(P_i)$ such that $\vec{f} \preceq \vec{c}^*$, and $(n_i, \vec{g}(P_i), F)$ is in *OPEN*. From the transitivity of the dominance relation, it follows that, for all $\vec{f}' \in F(P)$, there is at least some $\vec{f} \preceq \vec{c}^* \prec \vec{f}'$. Therefore, P can never be selected for expansion. □

THEOREM 4.6. *A dominated solution can never be selected for expansion.*

PROOF. Let us assume, for the purpose of contradiction, that a dominated solution $P = (s, \ldots, \gamma)$ with cost $\vec{g}_P$ is selected, that is, there exists $\vec{c}^* \in C^*$ with $\vec{c}^* \prec \vec{g}_P$. By definition, $H(\gamma) = \{\vec{0}\}$ and therefore $F(P) = \{\vec{g}_P\}$. However, by virtue of Lemma 4.5, such a path could never be selected for expansion. □

The following corollaries will be useful to compare the algorithm with MOA*.

COROLLARY 4.7. *At any time $COSTS \subseteq C^*$.*

PROOF. Trivial from Theorem 4.6. □

COROLLARY 4.8. *For each path P recorded in SG such that $\forall \vec{f} \in F(P)\, \exists \vec{c}^* \in C^*, \vec{c}^* \prec \vec{f}$, its extensions will never be recorded in SG.*

PROOF. By virtue of Lemma 4.5, such path could never be selected for expansion. □

THEOREM 4.9. *NAMOA* is admissible.*

PROOF. Let us assume a multiobjective search problem with solution. For any such problem NAMOA* might terminate or not. However, the latter possibility is ruled out by Theorem 4.3. If the algorithm terminates, it might terminate with or without solutions. Again, the latter possibility is ruled out by Theorem 4.2, since the algorithm only terminates when the OPEN list becomes empty, and this cannot happen until at least all nondominated solutions have been found. If the algorithm terminates with solutions, it might terminate with the set of all nondominated solutions or with some other set. Finally, Theorem 4.6 rules out the possibility of

NAMOA* finding other than nondominated solutions. Therefore, NAMOA* must always terminate with the set of all nondominated solutions. □

Theorem 4.9 shows NAMOA* is a fair competitor to MOA* and proves that the extra operations carried out by the latter can be safely skipped without compromising admissibility. Corollary 4.7 implies the algorithm could be terminated at any time with the guarantee that the set of solutions found so far are nondominated. Corollary 4.8 establishes a reasonable limit to the set of paths that will be recorded in *SG*. However, the example presented in Section 3.2 shows that Corollaries 4.7 and 4.8 do not apply to MOA*.

5. *Efficiency and Heuristics*

This section analyzes the effect of heuristic information on the efficiency of NAMOA* [Mandow and Pérez de la Cruz 2006]. The analysis parallels the argument presented for A* in Pearl [1984, pp. 79–85], though differences are needed to account for the nature of multiobjective preferences, and the fact that the new algorithm searches for *all* nondominated solutions. First, necessary and sufficient conditions for path expansion are developed. Then, NAMOA* algorithms are compared depending on their heuristic information, and the importance of the consistency property of heuristics is established. Finally, a comparison is drawn with the properties of A* and MOA*.

5.1. Necessary and Sufficient Conditions for Path Expansion.

The following definition is the multiobjective equivalent of the one presented for the scalar case [Pearl 1984, p. 80].

Definition 5.1. Let C be some set of cost vectors. We say that a path $P = (s = n_0, n_1, n_2, \ldots, n_k)$ is *C-bounded* with respect to $H(n)$ if for all subpaths $P_i = (n_0, n_1, \ldots, n_i)$ of P the following holds:

$$\exists \vec{h} \in H(n_i) \mid \nexists \vec{c} \in C, \quad \vec{c} \prec \vec{g}(P_i) + \vec{h}$$

An equivalent condition is,

$$\exists \vec{h} \in H(n_i) \mid \forall \vec{c} \in C \quad \vec{g}(P_i) + \vec{h} \precsim \vec{c} \ \vee \ \vec{g}(P_i) + \vec{h} = \vec{c}$$

When it becomes necessary to identify the heuristic function we shall use the notation C(H)-bounded.

Note that, by definition, a C^*-bounded path can never be filtered. Once such paths enter the *OPEN* list their fate is either to be pruned or selected for expansion.

THEOREM 5.2. *A sufficient condition for NAMOA* to select a path $P = (s, \ldots, n)$ for expansion is that: (a) P be a nondominated path to n; (b) P be C^*-bounded.*

PROOF. Let us assume, for the purpose of contradiction, that NAMOA* terminates and $P = (s = n_o, n_1 \ldots, n_k = n)$ was not selected. Since P is C^*-bounded, its subpaths can never be filtered. Since P is nondominated, its subpaths can never be pruned. Therefore, there will always be an open subpath $P_i = (n_0, \ldots, n_i)$ of P before termination. However, the algorithm can only terminate when *OPEN* becomes empty, and all subpaths of P and P itself will have to be selected. □

THEOREM 5.3. *A necessary condition for NAMOA* to select a path P for expansion is that P be C^*-bounded.*

PROOF. For a path $P = (s = n_0, n_1, n_2, \ldots, n_k)$ to be selected it must be in *OPEN* and also, by virtue of lemma 4.5,

$$\exists \vec{h} \in H(n) \quad | \quad \nexists \vec{c}^* \in C^*, \quad \vec{c}^* \prec \vec{g}(P) + \vec{h}$$

Now, if P is in *OPEN* then all its subpaths $P_i = (n_0, n_1, \ldots, n_i)$ must have been in *OPEN* and were selected for expansion. At that time each subpath must have satisfied Lemma 4.5. In summary, all subpaths P_i of P satisfy the following,

$$\exists \vec{h} \in H(n_i) \quad | \quad \nexists \vec{c}^* \in C^* \quad \vec{c}^* \prec \vec{g}(P_i) + \vec{h}$$

which is the definition of C^*-bounded path (Definition 5.1). □

Theorems 5.2 and 5.3 introduce a fundamental classification of paths in the graph. Those that are not C^*-bounded will never be considered for expansion. Among those C^*-bounded, nondominated ones will always be considered, while the fate of dominated ones is unclear: in the worst case, all of them will be considered, and in the best case none of them.

5.2. COMPARISON OF HEURISTICS. The necessary and sufficient conditions developed in the previous subsection are used here to give a broad characterization of when one heuristic function is more convenient than another. The intuition is that, in general, the greater the heuristic estimates, the smaller the set of C^*-bounded paths. The following definition introduces the notion of a more informed heuristic.[2]

Definition 5.4. A heuristic function $H_2(n)$ is said to be *at least as informed* as another $H_1(n)$ when both are admissible and all nodes n in the graph satisfy that,

$$\forall \vec{h}_2 \in H_2(n) \quad \exists \vec{h}_1 \in H_1(n) \quad | \quad \vec{h}_1 \preceq \vec{h}_2$$

THEOREM 5.5. *Let $H_1(n)$ and $H_2(n)$ be two admissible heuristics for the same problem. Let NAMOA$_1$ and NAMOA$_2$ be two versions of algorithm NAMOA* that differ only in the use of heuristic functions $H_1(n)$ and $H_2(n)$ respectively. If $H_2(n)$ is at least as informed as $H_1(n)$, then all nondominated and $C^*(H_2)$-bounded paths selected for expansion by NAMOA$_2$ will also be selected by NAMOA$_1$.*

PROOF. It suffices to show that if a path is $C^*(H_2)$-bounded, then it will also be $C^*(H_1)$-bounded. Since paths are also nondominated, application of Theorem 5.2, proves the paths will also be expanded by NAMOA$_1$.

A path $P = (s = n_0, n_1, n_2, \ldots, n_k)$ is C^*-bounded according to $H_2(n)$ if every subpath $P_i = (n_0, n_1, \ldots, n_i)$ satisfies (Definition 5.1),

$$\exists \vec{h}_2 \in H_2(n_i) \quad | \quad \forall \vec{c}^* \in C^* \quad \vec{g}(P_i) + \vec{h}_2 \precsim \vec{c}^* \vee \ \vec{g}(P_i) + \vec{h}_2 = \vec{c}^*$$

Now, since H_2 is at least as informed as H_1, for all $\vec{h}_2 \in H_2$ there exists a vector $\vec{h}_1 \in H_1$ such that $\vec{h}_1 \preceq \vec{h}_2$ (Definition 5.4). Therefore, every subpath P_i will satisfy one of,

[2] Note there is some erratum in the analogous definition in Stewart and White [1991].

—$\vec{g}(P_i) + \vec{h}_1 \preceq \vec{g}(P_i) + \vec{h}_2 = \vec{c}^* \Rightarrow \vec{g}(P_i) + \vec{h}_1 \preceq \vec{c}^*$
—$\vec{g}(P_i) + \vec{h}_1 \preceq \vec{g}(P_i) + \vec{h}_2 \precsim \vec{c}^*$

Since $\prec$ is transitive, and all natural q and all vectors $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^q$ satisfy the relationship $\vec{x} \prec \vec{y} \wedge \vec{y} \sim \vec{z} \Rightarrow \vec{x} \precsim \vec{z}$, it is possible to conclude that, $\vec{g}(P_i) + \vec{h}_1 \precsim \vec{c}^*$.

In consequence, P is also $C^*(H_1)$-bounded. □

Regrettably, the implications of Theorem 5.5 are limited. In general, it is not possible to determine which dominated C^*-bounded paths will be expanded by each version of the algorithm.

5.3. CONSISTENCY AND MONOTONICITY. We are now ready to show that consistency plays an important role in the efficiency of NAMOA*. The following definitions are equivalent to those presented in Stewart and White [1991, p. 806], and extend the scalar definitions of consistency and monotonicity already presented in Section 2.1.

Definition 5.6. A multiobjective heuristic function $H(n)$ is *consistent* if for all pairs of nodes n, n' in the graph, for all nondominated paths between them $P = (n, \ldots, n')$, and for all heuristic cost vectors $\vec{h}' \in H(n')$, the following condition holds,

$$\exists \vec{h} \in H(n) \quad | \quad \vec{h} \preceq \vec{c}(P) + \vec{h}'$$

Definition 5.7. A multiobjective heuristic function $H(n)$ is *monotone* if for all arcs (n, n') in the graph, the following condition holds,

$$\forall \vec{h}' \in H(n') \quad \exists \vec{h} \in H(n) \quad | \quad \vec{h} \preceq \vec{c}(n, n') + \vec{h}'$$

It can be shown that $H(n)$ is consistent if and only if it is monotone [Stewart and White 1991, Lemma 18], and that if $H(n)$ is monotone, then it is also admissible [Stewart and White 1991, Lemma 18].

The following theorem introduces a new necessary condition for path expansion when heuristics are monotone.

THEOREM 5.8. *If the heuristic function $H(n)$ is monotone, a necessary condition for NAMOA* to select a path $P = (s, \ldots, n)$ for expansion is that P be a nondominated path to n.*

PROOF. Let us assume, for the purpose of contradiction, that NAMOA* selects a path $P' = (s, \ldots, n)$ dominated to n, and represented in *OPEN* by the tuple $(n, \vec{g}(P'), F_{P'})$. There must be some path $P^* = (s = n_0, n_1, \ldots, n_k = n)$ yet to be discovered that is nondominated to n and dominates P'.

Since P^* is nondominated its subpaths will never be pruned. Since $\vec{g}(P^*) \prec \vec{g}(P')$ and the heuristic is monotone, if P' was not filtered, none of the subpaths of P^* will have been filtered either. Therefore, there must be some subpath $P_i^* = (n_0, n_1, \ldots, n_i)$ of P^* in *OPEN*.

Let $P_{i2}^* = (n_i, \ldots, n)$ be the subpath of P^* from n_i to n. Now, $\vec{g}(P^*) = \vec{g}(P_i^*) + \vec{g}(P_{i2}^*)$ and since costs are nonnegative and the relationship $\prec$ is transitive,

we have that,

$$\vec{g}(P_i^*) \preceq \vec{g}(P^*) \prec \vec{g}(P') \quad \Rightarrow \quad \vec{g}(P_i^*) \prec \vec{g}(P')$$

From the definition of consistency,

$$\forall \vec{h} \in H(n) \quad \exists \vec{h}' \in H(n_i) \quad | \quad \vec{h}' \preceq \vec{g}(P_{i2}^*) + \vec{h}$$

Combining both results,

$$\forall \vec{f} \in F(P') \quad \exists \vec{f}' \in F(P_i^*) \quad | \quad \vec{f}' \prec \vec{f}$$

Therefore, P' could never be selected from *OPEN* until all subpaths P_i^* of P had been selected. However, if $P_i^* = P^*$ ever entered *OPEN*, P' would be pruned. □

The results presented in previous subsections can be now restated for the case of consistent (monotone) heuristics.

THEOREM 5.9. *If H is consistent, then a necessary and sufficient condition for NAMOA to select some path $P = (s, \ldots, n)$ for expansion is that: (a) P be nondominated to n; (b) P be C^*-bounded.*

PROOF. Theorems 5.3 and 5.8 establish as necessary conditions for path expansion the same sufficient conditions established by Theorem 5.2. □

THEOREM 5.10. *Let $H_1(n)$ and $H_2(n)$ be two admissible heuristics for the same problem. Let $H_2(n)$ be additionally consistent. Let NAMOA$_1$ and NAMOA$_2$ be two versions of NAMOA* that differ only in the use of heuristic functions $H_1(n)$ and $H_2(n)$ respectively. If $H_2(n)$ is at least as informed as $H_1(n)$, then all paths selected for expansion by NAMOA$_2$ will also be selected for expansion by NAMOA$_1$.*

PROOF. Obvious from Theorems 5.5 and 5.8, since all paths selected by NAMOA$_2$ will be nondominated □

5.4. DISCUSSION. The importance of consistency established by Theorem 5.10 in the efficiency of NAMOA* is analogous to that established for the efficiency of A* [Pearl 1984]. Consistency guarantees that only the strictly necessary C^*-bounded paths will be selected for expansion. In other words, the pruning of C^*-bounded paths will be optimal. Additionally, if a consistent heuristic H_2 is used, and H_2 is more informed than another heuristic H_1 then, for all problem instances, the set of paths selected with H_2 will always be a subset of those selected with H_1. In particular, under Definition 5.4, any consistent heuristic will be more informed than a blind search, that is, $H(n) = \{\vec{0}\}\ \forall n$.

These results are in sharp contrast to those obtained for MOA*. A comparison of heuristic functions could be established in terms of the set of nodes selected for expansion [Stewart and White 1991]. However, no clear benefits on search efficiency could be established, since "the significance of this comparative measure varies widely among different problem domains" (p. 802). The properties of consistency and monotonicity did not provide much advantage since they were found "not sufficient to guarantee that nodes will not need to be reopened" (p. 806). This led to the conclusion that "the properties of monotonicity and consistency are not as important [in MOA*] as they are in single objective search" (p. 812).

This is a consequence of the *node* expansion operation that characterizes MOA*, that is, each time a node is selected for expansion, all paths reaching that node in the search graph are extended simultaneously. MOA* will expand each node reached by *at least one* C^*-bounded path [Stewart and White 1991, Lemma 17]. In general, *all* paths reaching each node upon expansion are not necessarily C^*-bounded, but will be expanded as well. When the heuristic is consistent, MOA* guarantees that a nondominated path reaches each node upon expansion [Stewart and White 1991, Lemma 20]. However, there is no guarantee that *all* paths reaching the node at that time are also nondominated. The example presented in Section 3.2 showed that MOA* may expand dominated paths even when the heuristic is consistent, and that nodes may, in fact, be reexpanded unnecessarily. Note that the heuristic function presented in Table II trivially satisfies the consistency property. NAMOA* avoids these problems expanding paths individually under the guarantees established by Theorems 5.3, 5.2, and 5.9.

The consistency property plays a fundamental role in establishing the optimality of A* over the class of admissible algorithms. The question arises whether NAMOA* might be proven optimal among the class of admissible multiobjective search algorithms under analogous conditions. The next section takes up this question.

6. *Optimality of NAMOA**

The optimality of A*, in the sense of being the most computationally efficient algorithm among those equally informed (i.e., with access to the same heuristic information), was a subject of debate almost from its inception. The computational efficiency of A* has been traditionally established in terms of the set of expanded nodes, and the number of expansion operations (iterations), for space and time complexity respectively. Concerning the set of all conceivable problem instances with admissible heuristics, and the class of admissible algorithms, Mérõ [1984] showed that no optimal algorithm exists when complexity is measured by the number of expansion operations. Dechter and Pearl [1985] further showed that no optimal algorithm exists even when complexity is measured by the number of *distinct* nodes expanded. Nevertheless, they showed that A* was optimal over admissible algorithms by both measures when applied to problems with consistent heuristics. In other words, no algorithm equipped only with the same consistent heuristics can skip a single expansion operation performed by A* without compromising its admissibility.

This section extends these results to the multiobjective search framework. It is important to note that the "set of expanded nodes" is no longer a significant efficiency measure in multiobjective search [Stewart and White 1991, p. 802]. This is due to the fact that many different nondominated paths may reach each node in the search graph. With increasing depth, an exponentially large set of paths may reach each node in the worst case. It can be shown that the average number of non-dominated q-dimensional vectors in a set of size L is $O((log|L|)^{q-1})$, provided all $(n!)^q$ relative orderings are equally probable [Bentley et al. 1978]. Therefore, it is the number of cost vectors stored in the search graph that dominates space requirements in the general case. Since each cost vector in the graph corresponds to a different partial solution path, the following analysis is carried out in terms of the number of *path expansions* performed by the algorithm.

6.1. TERMINOLOGY AND ASSUMPTIONS. The following terminology and assumptions are borrowed or adapted to the multiobjective case from their scalar counterparts [Dechter and Pearl 1985].

The class of admissible multiobjective search algorithms will be denoted $\mathbf{M_{ad}}$. Additionally, we assume that each algorithm compared with NAMOA* uses the primitive computational step of *path expansion*, that it only expands paths previously generated, and that all paths emanate from the start node s. This explicitly excludes other strategies like bidirectional search.

We represent each problem instance by the quadruple $I = (G, s, \Gamma, H)$, that is, we consider the heuristic evaluation function $H(n)$ as part of the parameters that specify problem instances. We define the following sets of problem instances:

$$\mathbf{I_{MAD}} = \{(G, s, \Gamma, H) \mid \text{H is admissible}\}$$

$$\mathbf{I_{MCON}} = \{(G, s, \Gamma, H) \mid \text{H is consistent}\}.$$

Since consistency implies admissibility $\mathbf{I_{MCON}} \subset \mathbf{I_{MAD}}$.

We adopt the following definitions of dominance and optimality of algorithms.

Definition 6.1. Algorithm A is said to *dominate* algorithm B relative to a set $\mathbf{I}$ of problem instances iff in every instance $I \in \mathbf{I}$, the set of paths expanded by A is a subset of the set of paths expanded by B. A *strictly dominates* B iff A dominates B and B does not dominate A.

Definition 6.2. Algorithm A is optimal over a class $\mathbf{A}$ of algorithms iff A dominates every member of $\mathbf{A}$. Algorithm A is nondominated over $\mathbf{A}$ if no member of $\mathbf{A}$ dominates A.

We shall denote by $\mathbf{P^{C*}}$the set of paths definitely expanded by NAMOA* (see Theorems 5.2 and 5.9), and by $P^*(n, m)$ the set of nondominated paths from n to m in a given graph G.

Finally, we shall find it useful to introduce the following relaxed sufficient condition for admissibility.

Definition 6.3. A heuristic function H satisfies the relaxed sufficient condition for admissibility if,

$$\forall n \in G \quad \forall \gamma \in \Gamma \quad \forall P_{n\gamma} \in P^*(n, \gamma) \quad \exists \vec{h} \in H(n) \quad | \quad \vec{h} \preceq \vec{g}(P_{n\gamma})$$

The definition of admissible heuristic function (Assumption 2, Section 4) required this to hold only for all pairs of nodes (n, γ) such that $\gamma \in \Gamma$ and a nondominated solution reaches γ through n. Extending the condition to all pairs of nodes (n, γ) such that $n \in G$ and $\gamma \in \Gamma$ trivially implies admissibility.

6.2. OPTIMALITY WITH CONSISTENT HEURISTICS

THEOREM 6.4. *Any algorithm in* $\mathbf{M_{ad}}$ *expands in each* $I \in \mathbf{I_{MCON}}$ *all paths in* $\mathbf{P^{C*}}$.

PROOF. Let $I \in \mathbf{I_{MCON}}$ be an instance where some algorithm $\mathrm{B} \in \mathbf{M_{ad}}$ avoids expanding a path $P_{sn'} \in \mathbf{P^{C*}}$. We shall show, for the purpose of contradiction, that a new problem instance $I_2 \in \mathbf{I_{MAD}}$ can always be built where B cannot find all nondominated solutions, that is, $\mathrm{B} \notin \mathbf{M_{ad}}$.

The new problem instance $I_2 = (G_2, s, \Gamma_2, H_2)$ will be a simple extension of $I = (G, s, \Gamma, H)$. Our goal is to include in I_2 a new extension of path $P_{sn'} = (s, \ldots, n')$, such that, the original heuristic function H remains admissible, and the new extension is a new nondominated solution. In such new problem instance Algorithm B would behave as in instance I, avoiding expansion of $P_{sn'}$ and therefore skipping the new solution.

Let us assume that $H(n') = \{\vec{0}\}$. Then, I_2 can be built from I simply setting n' as a new goal node, that is, $G_2 = G$, $H_2 = H$, and $\Gamma_2 = \Gamma \cup \{n'\}$.

H_2 trivially satisfies the admissibility condition for all goal nodes in Γ. We shall show that it also satisfies the relaxed sufficient condition for admissibility (Definition 6.3) for all pairs of nodes (m, n'), $m \in G$, that is,

$$\forall m \in G \quad \forall P_{mn'} \in P^*(m, n') \quad \exists \vec{h} \in H(m) \quad | \quad \vec{h} \preceq \vec{g}(P_{mn'})$$

From the consistency of H (Definition 5.6, Section 5.3), we know that,

$$\forall m \in G \ \forall P_{mn'} \in P^*(m, n') \quad \forall \vec{h}'_n \in H(n') \quad \exists \vec{h}_m \in H(m) \quad | \quad \vec{h}_m \preceq \vec{g}(P_{mn'}) + \vec{h}'_n$$

Combining this definition with the assumption that $H(n') = \{\vec{0}\}$, we obtain the desired result.

In order to ruin the admissibility of algorithm B in I_2, it suffices to show that $P_{sn'}$ is a *new* nondominated solution in I_2. Since NAMOA* selected $P_{sn'}$ for expansion in I, it must be C*-bounded (Theorem 5.9), that is,

$$\exists \vec{h} \in H(n') \quad | \quad \forall c^* \in C^* \quad \vec{g}(P_{sn'}) + \vec{h} \preceq \vec{c}^* \ \vee \ \vec{g}(P_{sn'}) + \vec{h} = \vec{c}^*$$

Combining this definition with the assumption that $H(n') = \{\vec{0}\}$, we obtain,

$$\forall c^* \in C^* \quad \vec{g}(P_{sn'}) \preceq \vec{c}^* \ \vee \ \vec{g}(P_{sn'}) = \vec{c}^*$$

which, by definition, implies $P_{sn'}$ is a nondominated solution in I_2.

Let us assume now the more general case where $H(n') \neq \{\vec{0}\}$. Then, I_2 can be built in the following way. For each heuristic cost vector $\vec{h}_i \in H(n')$, create a new node t_i and a new arc from n' to t_i with cost vector $\vec{c}(n', t_i) = \vec{h}_i$. Let Γ' denote the set of such new nodes. All new nodes will be set as new goal nodes in I_2, that is, $\Gamma_2 = \Gamma \cup \Gamma'$. H_2 is set equal to H except for the new nodes, where $H_2(t_i) = \{\vec{0}\}$ $\forall t_i \in \Gamma'$ (see Figures 9 and 10).

H_2 trivially satisfies the admissibility condition for all goal nodes in Γ. We shall show that it also satisfies the relaxed sufficient condition for admissibility (Definition 6.3) for all pairs of nodes (m, t_i), $t_i \in \Gamma'$.

From the consistency of H,

$$\forall m \in G \ \forall P_{mn'} \in P^*(m, n') \ \forall \vec{h}_i \in H(n') \quad \exists \vec{h}_m \in H(m) \quad | \quad \vec{h}_m \preceq \vec{g}(P_{mn'}) + \vec{h}_i$$

Susbstituting the fact that $\forall t_i \in \Gamma' \ \vec{c}(n', t_i) = \vec{h}_i$ and $\vec{g}(P_{mt_i}) = \vec{g}(P_{mn'}) + \vec{c}(n', t_i)$ and, since all paths reaching each t_i are extensions of paths reaching n' we obtain the desired result,

$$\forall m \in G \quad \forall t_i \in \Gamma' \quad \forall P_{mt_i} \in P^*(m, t_i) \quad \exists \vec{h} \in H(m) \quad | \quad \vec{h} \preceq \vec{g}(P_{mt_i})$$

Therefore, H_2 is admissible in I_2.

Once again, to ruin the admissibility of algorithm B in I_2 it suffices to show that the cost of at least one of the newly created solution paths is nondominated in I_2.

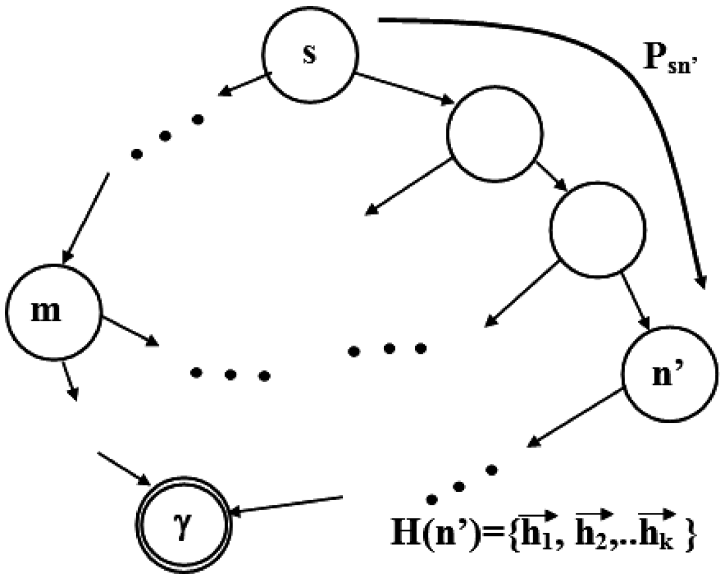


FIG. 9. A problem where algorithm B avoids expansion of $P_{sn'}$.

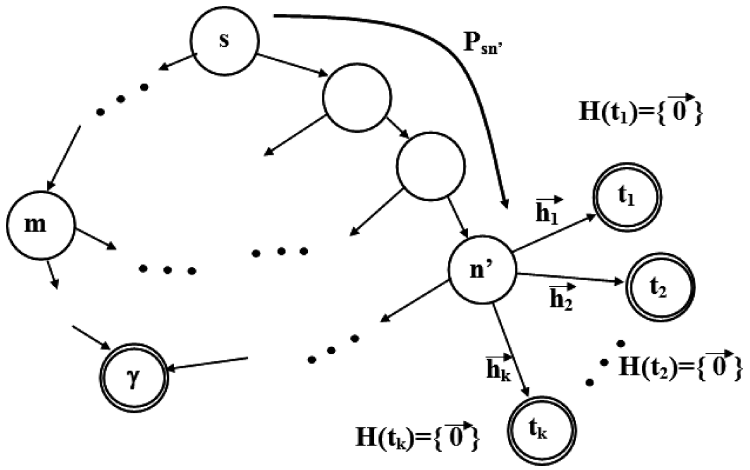


FIG. 10. A problem where algorithm B cannot find all nondominated solutions.

Since NAMOA* selected $P_{sn'}$ for expansion in I, it must be C*-bounded, that is,

$$\exists \vec{h}_b \in H(n') \quad | \quad \forall c^* \in C^* \quad \vec{g}(P_{sn'}) + \vec{h}_b \preceq \vec{c}^* \vee \vec{g}(P_{sn'}) + \vec{h}_b = \vec{c}^*$$

Note that for some new node $t_b \in \Gamma'$ its cost is $\vec{g}(P_{st_b}) = \vec{g}(P_{sn'}) + \vec{h}_b$. Therefore, by definition, at least the newly created path to t_b is nondominated. □

Theorem 6.4 establishes the optimality of NAMOA* among admissible multiobjective algorithms over problems with consistent heuristics, when efficiency is measured by the number of path expansion operations. No algorithm can skip a path expanded by NAMOA* in $\mathbf{I_{MCON}}$ without compromising its admissibility. Regarding MOA*, this means the example presented in Section 3.2 and discussed

in Section 3.3 is not a singular exception. NAMOA* strictly dominates MOA* over $\mathbf{I_{MCON}}$ and will always expand a subset of the paths considered by MOA* over this problem set.

6.3. OPTIMALITY AND NONCONSISTENT HEURISTICS. An analysis of optimality in $\mathbf{M_{ad}}$ over $\mathbf{I_{MAD}}$ is well beyond the scope of this article. Nevertheless, this section draws upon previous relevant work to discuss the following two propositions:

(1) There is no optimal algorithm in $\mathbf{M_{ad}}$ over $\mathbf{I_{MAD}}$.

(2) NAMOA* is *not* even nondominated in $\mathbf{M_{ad}}$ over $\mathbf{I_{MAD}}$.

The absence of an optimal algorithm in $\mathbf{M_{ad}}$ over $\mathbf{I_{MAD}}$ is quite straightforward from previous results in scalar search. As stated above, it is well known that, for scalar admissible algorithms, there is no optimal algorithm for the set of search problems with admissible heuristics. The best we can hope for is a nondominated algorithm, that is, one that beats all other algorithms in some problem instances, but is beaten by some of them in others. Scalar algorithms that search for *all* optimal solutions are a particular case of multiobjective ones, that is, the case where the number of objectives is one. Mérõ [1984] presented a set of sample problems with nonconsistent heuristics where no admissible algorithm could perform optimally in the set of paths expanded. Since each of these problems had a single solution, the counterexample is also valid for the class of scalar algorithms that searches for all solutions as well and, for that matter, for all its superclasses including $\mathbf{M_{ad}}$.

Going back to scalar search, several alternatives have been proposed to improve A* for the case of nonconsistent heuristics [Mahanti and Ray 1988]. In particular, Mérõ [1984] introduced the idea of propagating heuristic values to the descendants of every node. This technique, known as *pathmax* in the literature, amounts to using the following evaluation function,

$$f(n) = max\{g(n') + h(n') \mid \text{n' on the current path to n}\} \tag{6}$$

Dasgupta et al. [1995] extended this idea to the multiobjective search framework and showed that introducing pathmax in partial order best-first search algorithms retains admissibility. They also showed that if a multiobjective best-first Algorithm A is the pathmax version of Algorithm B, then A dominates B both in the sets of nodes and paths expanded.

Let us denote by NAMOA** a version of NAMOA* that incorporates the partial order pathmax strategy. It is obvious from these results that NAMOA** dominates NAMOA* in $\mathbf{M_{ad}}$ over $\mathbf{I_{MAD}}$. However, whether NAMOA** itself is nondominated on this class remains an open research question.

Dechter and Pearl [1985] introduced A**, the pathmax version of A*, and showed that A** was nondominated among admissible algorithms over admissible problem instances. However, they used the set of expanded *nodes* as a comparative measure. No algorithm has been proven nondominated in this class with respect to the set of expanded *paths*. Mahanti and Ray [1988] have pointed out that a question of much greater interest would be to find an algorithm with good average running time behavior. This also still remains an open research question both in the scalar and the multiobjective search domains.

7. *Conclusions and Future Work*

The paper describes and analyzes NAMOA*, an extension of A* to multiobjective search problems. Like A*, the algorithm uses the basic operations of path selection and expansion. The new algorithm was compared to MOA*, a previous proposal, and has been shown to require fewer operations. NAMOA* is admissible under reasonable assumptions and presents important properties, analogous to those of A*, that are not shared by MOA*. In particular, bounds can be established on the number of paths that will be selected for expansion and stored in the search graph. The efficiency of NAMOA*, measured by the set of expanded paths, improves with heuristic information whenever heuristics are consistent. NAMOA* has also been proven optimal among admissible multiobjective algorithms over the class of problems with consistent heuristics. In this sense, NAMOA* was found to strictly dominate MOA*. In summary, the most important beneficial properties of A* have been shown to prevail in NAMOA*.

There are a number of open research issues in multiobjective search. Previous results imply that no optimal admissible multiobjective algorithm exists over the class of problems with general admissible heuristics when the set of expanded paths is used as a comparative measure. NAMOA* is not even nondominated over this class. Determining nondominated algorithms over this class is an important area of future research.

Multiobjective search is known to have exponential space and time requirements with graph size in the worst case, even with just two objectives [Hansen 1979]. However, search in polynomial state spaces with bounded integer costs can be shown to require only polynomial resources with solution depth, although still exponential with the number of objectives [Mandow and Pérez de la Cruz 2009]. A deeper analysis of the complexity of multiobjective search is also an important research issue.

Finally, there are a number of practical issues related to the efficiency of multiobjective search that deserve to be investigated. There are a number of implementation details and options open in the general algorithm described. In particular, the relation of the number of path selection operations with real space and time requirements in potential application areas should be empirically analyzed.

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